

Minimal block I: the log-weighted primitive and the exact product-density formula

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Abstract

The divide-and-integrate tower on a primitive $G(x) = \int_0^x g$ is the log-weighted integral $H_j(y) = \frac{1}{j!} \int_0^y \log^j(y/t) g(t) dt$ (Fubini on the triangle $0 < s \leq t \leq y$). For the weighted kernel $g(t) = t^{p-1} e^{-\beta t^2 + \beta a t}$ this turns the exact tower reduction of unit 50 into the product-density formula for an equal-exponent chart block, $Z_{d+1}(c) = (\prod_{i \leq d} k_i^{-1}) c^{-p} \frac{1}{d!} \int_0^{cB} \log^d(cB/t) t^{p-1} e^{-\beta t^2 + \beta a t} dt$, valid at every scale $c > 0$. Integrability of $|\log t|^{i\gamma} e^{-\beta t^2 + \beta a t}$ on $(0, \infty)$ is recorded for the domination arguments to come. Zero sorry/axiom.

1 The things formalised

```
theorem integral_Icc_log_pow_div (j : ℕ) (s y : ℝ) (hs : 0 < s) (hsy : s ≤ y) :
  t in Icc s y, (Real.log t - Real.log s) ^ j / t
  = (Real.log y - Real.log s) ^ (j + 1) / (j + 1)
noncomputable def logWeightedPrim (g : ℝ → ℝ) (j : ℕ) (y : ℝ) : ℝ :=
  1 / (j.factorial : ℝ) * t in Ioc (0 : ℝ) y, (Real.log y - Real.log t) ^ j * g t
theorem iterDivPrim_primitive (g) (hg : Measurable g) (hint : ...) (j : ℕ) (y : ℝ)
  (hy : 0 < y) :
  iterDivPrim (fun x => t in Ioc (0 : ℝ) x, g t) j y = logWeightedPrim g j y
theorem logPow_weightedKernel_integrableOn (a : ℝ) (h : 0 < ) (h : -1 < ) (i : ℕ) :
  IntegrableOn (fun t => |Real.log t| ^ i * (t ^ a * quadKernel a t)) (Ioi 0)
theorem iterChartGen_eq_logWeighted ...
  (hp : i, i → d → ((h i : ℝ) + 1) / k i = p) (hc : 0 < c) :
  iterChartGen a b k h (d + 1) c
  = ( i Finset.range (d + 1), (1 : ℝ) / k i) * c ^ (-p)
  * logWeightedPrim (fun t => t ^ (p - 1) * quadKernel a t) d
  (c * b ^ ( i Finset.range (d + 1), k i))
```

2 Proof strategy

The triangle integrand $f(t, s) = \mathbf{1}_{s \leq t} (\log t - \log s)^j g(s)/t$ is measurable (`Measurable.ite` on $\{s \leq t\}$). Its s -section over $(0, y]$ is the inner integral over $(0, t]$ (`setIntegral_indicator` with `Iic t`); its t -section is $g(s) \int_s^y (\log t - \log s)^j / t dt = g(s) \log^{j+1}(y/s)/(j+1)$ by the fundamental theorem of calculus. `integrable_prod_iff'` reduces integrability on the square to that of $\log^{j+1}(y/s)|g(s)|$, and `integral_integral_swap` finishes the induction. For the kernel, $|\log t| \leq t^{-\varepsilon}/\varepsilon$ on $(0, 1]$ and $|\log t| \leq t$ on $[1, \infty)$ dominate $|\log t|^{i\gamma}$ by integrable weighted kernels; the binomial expansion of $(|\log y| + |\log t|)^j$ gives the $(0, y]$ version.

3 Roadblocks and resolutions

- `positivity` cannot see $0 \leq t^\gamma \text{quadKernel}$ (opaque definition): derive the fact once from `quadKernel_pos` and thread it.
- `add_pow` expands $(x + y)^n$ as $\sum x^m y^{n-m} \binom{n}{m}$; the dominating sum has to be written in that orientation.
- After `congr 1` on set integrals the goal may be a measure equality; rewrite the set identity before, not after.
- `Integrable.congr` hands the pointwise goal as a beta redex; use `simp only` rather than `rw`.

4 Outcome

Astra target 1 (product-density identity) is done in the form best adapted to the existing tower. Next: the frozen block at general scale (pointwise limit $N^p I(N\gamma) / \log^{m-1} N \rightarrow \gamma^{-p} A_{p-1}(a) / ((m-1)! \prod k_i)$ and the uniform envelope), then the noncritical integration and the freezing argument.